

1. GENERAL

- 1.1 TITLE: **Sports Biomechanics**
1.2 COURSE NUMBER: BM:321
1.3 CREDITS: 3L-0T-0P- Credits-9
1.4 SEMESTER-OFFERED: ODD
1.5 PRE-REQUISITES: Engineering Mechanics, Biomechanics.

2. OBJECTIVE: The purpose of the course is to supply information and understanding of the biomechanics of human movement, performance, training, injury, and prevention, etc. Evaluate movement technique using a movement analysis model. Evaluate the mechanics of exercises and activities as they affect the human body. Performance analysis of external devices used for activities of daily living, exercise and sport. Apply principles related to internal tissue loading to improving tissue structure and function.

3. COURSE TOPICS:

Unit1: Introduction to Sports Biomechanics, Linear Kinematics, Angular Kinematics Equilibrium and Human Movement and gait, Centre of mass, Work, Power, Energy, Torques & Moments.

Unit 2: Kinematic and kinetic Concepts for analyzing Human Motion, Measure of kinematic and kinetic variables. The Biomechanics of Human Upper and lower Extremity, Estimation of anthropometric parameters and joint forces and muscular moments during an athletic movement. The use of videography in recording sports movements, Data Processing, Projectile motion.

Unit 3: Energy, work and muscular power during an athletic movement, The motor function and the breaking function of the muscle, Classification of the exercises, Transfer and transformation of energy, Utilization of the elastic properties of biological structures.

Unit 4: The mechanisms of terrestrial locomotion, The mechanics of walking and running, The mechanics of speed skating, The mechanics of bicycling, Efficiency of terrestrial locomotion.

Unit 5: The mechanics of swimming, The mechanics of rowing, Efficiency of aquatic locomotion.

Unit 6: Analysis of athletic movements, The rotation movements in gymnastic, The mechanics of sprint running, The mechanics of throwing in athletics.

4. READINGS

4.1 TEXTBOOK:

1. Bartlett R., (2002) Introduction to Sports Biomechanics. Second Ed. Taylor & Francis, New York, NY10017, USA.
2. Knudson D., (2007) Fundamentals of Biomechanics. Second Ed. Springer, New York, USA.
3. Huston R. L., (2010) Principles of Biomechanics. CRC Press Taylor & Francis Group, Sound ParkwayNW, Boca Raton, FL.

4.2 REFERENCE BOOKS:

1. Susan J. H., (2006) Basic Biomechanics, Fifth Ed. McGrawHill.
2. Watkins, J (2007) An introduction to biomechanics of sport and exercise, Church Livingstone Elsevier.
3. Nigg, B. M., & Herzog, W. (2007) Biomechanics of the Musculo-skeletal System. Third Ed. John Wiley & Sons Ltd, West Sussex, England
4. Vaughn, C. L., Davis B. L., & Connor J. C.(1999) Dynamics of Human Gait . Second Ed. KibohoPublishers, Cape Town, South Africa.

6. MacLaren D., Reilly T., & Lees A., (2005) Biomechanics and Medicine in Swimming. Second Ed. Spon Press 11 New Fetter Lane London.